

STRUCTURAL STABILITY: THEORY AND IMPLEMENTATION

is a practical work that provides engineers and students in structural engineering or structural mechanics with the background needed to make the transition from fundamental theory to practical design rules and computer implementation. Beginning with the basic principles of structural stability and basic governing equations, **STRUCTURAL STABILITY** is a concise and comprehensive introduction that applies the principles and theory of structural stability (which are the basis for structural steel design) to the solution of practical building frame design problems. Special features include:

- modern theories of structural stability of members and frames, and a discussion of how these theories may be utilized to provide design rules and calculation techniques for design
- important governing equations and the classical solutions used in design processes

**Chen
Lui**

**STRUCTURAL
THEORY AND IMPLEMENTATION**

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PREFACE

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techniques suitable for design office use. In this way, the reader not only will obtain an understanding of the fundamental principles and theory of structural stability from an idealized elastic, perfect system, but also to an inelastic imperfect system that leads to the necessary links between the code rules, design office practice, and the actual structural system in the real world.

The continued rapid development in computer hardware and software in recent years has made it possible for engineers and designers to predict structural behavior quite accurately. The advancement in structural analysis techniques coupled with the increased understanding of structural behavior has made it possible for engineers to adopt the Limit States Design philosophy. A limit state is defined as a condition at which a structural system or its component ceases to perform its intended function under normal conditions (Serviceability Limit State) or failure under severe conditions (Ultimate Limit State). The recently published Load and Resistance Factor Design (LRFD) Specification for the

Professor H. L. Michael of Purdue University for continuing support over many years, and to the graduate students, C. Cheng, L. Duan, and F. H. Wu, among others, for preparing the Answers to Some Selected Problems during their course work on Structural Stability in the spring semester of 1986 in the School of Civil Engineering at Purdue University.

December, 1986
West Lafayette, IN

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NOTATION

M_{ext}	=	moment at a section due to externally applied loads
M_{int}	=	internal resisting moment of the section
M_{tr}	=	transition moment (in Plastic Design)
M_n	=	nominal flexural strength
M_p	=	ZF_y = plastic moment
M_{px}	=	$1.18M_{\text{px}} \left[1 - \left(\frac{P}{P_y} \right) \right] \leq M_{\text{px}}$ for H-section about strong axis.
	=	plastic moment capacity about the strong axis considering the influence of axial load
M_{py}	=	$1.19M_{\text{py}} \left[1 - \left(\frac{P}{P_y} \right)^2 \right] \leq M_{\text{py}}$ for H-section about weak axis.
	=	plastic moment capacity about the weak axis considering the influence of axial load

U_w	=	$\frac{1}{2} \int_0^L EC_w \left(\frac{d^2 y}{dz^2} \right)^2 dz$
	=	strain energy due to warping restraint torsion
V	=	$-W_{\text{ext}}$ = potential energy
W_{int}	=	$-U$ = work done by the internal resisting forces
W_{ext}	=	$-V$ = work done by the external applied forces
Π	=	$U + V$ = total potential energy

GEOMETRY AND DIMENSIONS

A	=	cross sectional area
b_t	=	flange width
C_w	=	warping constant
	=	$\frac{1}{2} I_y h^2$ for I section
d	=	depth

E_{eff}	=	effective modulus
E_r	=	reduced modulus
E_t	=	tangent modulus
F_y, σ_y	=	yield stress
G	=	shear modulus
	=	$\frac{E}{2(1 + \nu)} = 11,200 \text{ ksi for steel}$
ν	=	Poisson's ratio
	=	0.3 for steel

STABILITY RELATED FACTORS

A_F	=	amplification factor
B_1	=	$P - \delta$ moment amplification factor for beam-columns in LRFD

	=	$1 + \psi \frac{P}{P_{ek}} = \text{effective length factor}$
κ	=	$\sqrt{\frac{P_c}{P_{ek}}} = \text{effective length factor}$
r_i	=	load factors
ϕ	=	resistance factor
ϕ_b	=	resistance factor for flexure = 0.90
ϕ_c	=	resistance factor for compression = 0.85

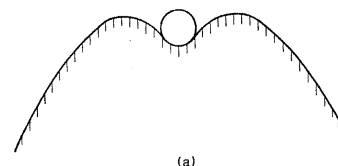
Chapter 1

GENERAL PRINCIPLES

change in geometry of the structure must be taken into account; as a consequence, equilibrium equations must be written based on the geometry of a structure that becomes deformed under load. This is known as the *second-order analysis*. The second-order analysis is further complicated by the fact that the resulting equilibrium equations are differential equations instead of the usual algebraic equations. Consequently, a mastery of differential calculus is a must before any attempt to solve these equations.

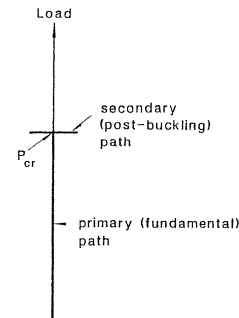
In what follows, we will explain the nature of structural stability and ways to analyze it accurately.

The concept of stability is best illustrated by the well-known example of a ball on a curved surface (Fig. 1.1). For a ball initially in equilibrium, a slight disturbing force applied to the ball on a concave surface (Fig. 1.1a) will displace the ball by a small amount, but the ball will return to its initial equilibrium position once it is no longer being disturbed. In this case, the ball is said to be in a *stable equilibrium*. If the disturbing force is

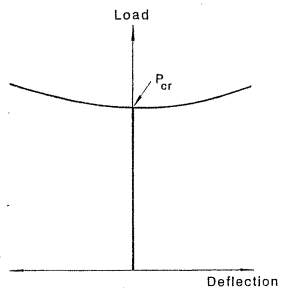


equilibrium or from a state of stable to unstable equilibrium is marked by the *stability limit point*. The tangent stiffness of the system vanishes at just this point. We shall use the principle of vanishing tangent stiffness to calculate the buckling load of a system in subsequent sections and chapters.

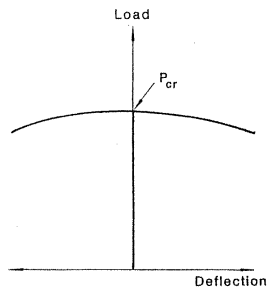
Stability of an elastic system can also be interpreted by means of the concept of *minimum total potential energy*. In nature, an elastic system always tends to go to a state in which the total potential energy is at a minimum. The system is in a stable equilibrium if any deviation from its initial equilibrium state will result in an increase in the total potential energy of the system. The system is in an unstable equilibrium if any deviation from its initial equilibrium state will result in a decrease in total potential energy. Finally, the system is in a neutral equilibrium if any deviation from its initial equilibrium state will produce neither an increase nor a decrease in its total potential energy. Because of this principle, the energy concept can be used to find the buckling load of a system.



General Principles

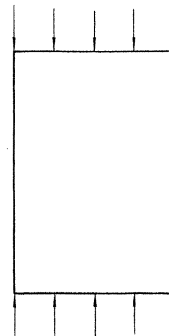


(a) STABLE SYMMETRIC BIFURCATION

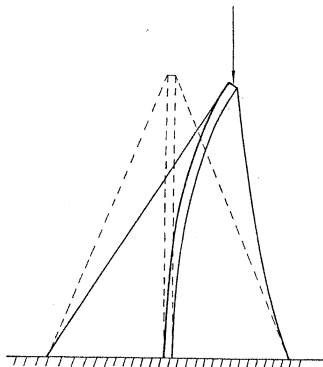


(b) UNSTABLE SYMMETRIC BIFURCATION

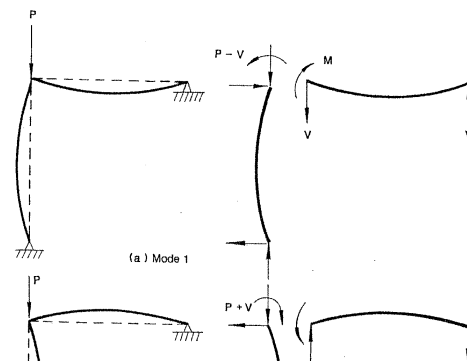
1.2 Types of Stability



General Principles



1.2 Types of Stability



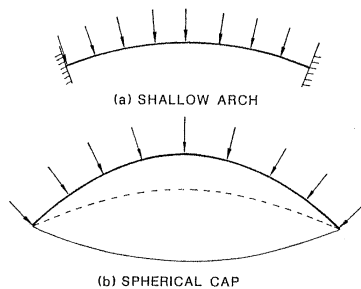


FIGURE 1.8 Examples of limit point buckling

1.3 METHODS OF ANALYSES IN STABILITY

The concept of stability as described in Section 1.1 can be used to determine the critical conditions of an elastic system that is susceptible to instability.

Bifurcation Approach

The first approach is called the *bifurcation approach*. In this approach, the state at which two or more different but adjacent equilibrium configurations can exist is sought by an *eigenvalue analysis*. The lowest load that corresponds to this state is the critical load of the system. At the critical load, equilibrium can be maintained with alternative deflection modes that are infinitesimally close to one another.

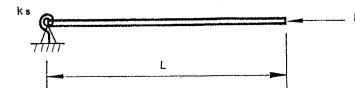
To determine the critical load using the bifurcation approach, it is necessary to identify all possible equilibrium configurations the system can assume at the bifurcation load. This can best be accomplished by

term “conservative forces” used here are those forces whose potential energy is dependent only on the final values of deflection, not the specific paths to reach these final values. If the system is in equilibrium, its total potential energy must be stationary. Thus, by setting the first derivative of the total potential energy function with respect to each generalized displacement equal to zero, we can identify the equilibrium conditions of the system. The critical load can then be calculated from the equilibrium equations.

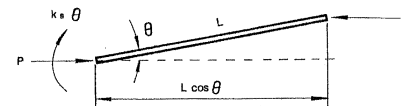
Please note that by setting the first derivative of the total potential energy function equal to zero, we can only identify the equilibrium conditions of the system. To determine whether the equilibrium is stable or unstable, we must investigate higher order derivatives of the total potential energy function.

Dynamic Approach

The critical load of an elastic system can also be obtained by the *dynamic*



(a)



value of k_s/L , the system will buckle. At this critical load, equilibrium for the bar is possible in both the original horizontal and slightly deflected positions.

Energy Approach

In using the energy approach to determine P_{cr} , one must write an expression for the total potential energy comprising the strain energy and the potential energy of the system. The strain energy stored in the system as the bar assumes its slightly deflected configuration is equal to the strain energy of the spring:

$$U = \frac{1}{2}k_s\theta^2 \quad (1.4.3)$$

The potential energy of the system is the potential energy of the external force and it is equal to the negative of the work done by the external force on the system:

determine not only the critical load, P_{cr} , but also the nature of the equilibrium of this system. That is, we can further determine whether the equilibrium of the system is stable or unstable at various stages of loading.

To determine whether the equilibrium is stable or unstable for the system in its original ($\theta = 0$) position, we need to investigate the positiveness or negativeness of higher derivatives of the total potential energy function. For this problem, if we take the second derivative of the total potential energy function given in Eq. (1.4.5), we have

$$\frac{d^2\Pi}{d\theta^2} = k_s - PL \cos \theta \approx k_s - PL \quad (1.4.9)$$

Thus, for $P < P_{cr}$, $d^2\Pi/d\theta^2$ is positive. This indicates that the equilibrium is *stable*. For $P > P_{cr}$, $d^2\Pi/d\theta^2$ is negative, and the equilibrium condition is therefore *unstable*. This behavior is illustrated in Fig. 1.11, in which the

small θ analysis and so no conclusion can be drawn. However, as will be demonstrated in the next section, the stability condition at the critical load ($P = P_{cr}$) can be investigated using the energy approach with a large θ analysis.

1.4.2 Rigid Bar Supported by a Translational Spring

Figure 1.12a shows a rigid bar hinged at one end and supported by a linear translational spring of stiffness k_s at the free end. The spring is assumed to be able to move freely in the horizontal direction but retains its vertical orientation as the bar deflects. The bar is subjected to a concentrated force P at the free end. Assuming the system is geometrically perfect, we shall determine the critical load of the system.

Bifurcation Approach

In Fig. 1.12b we see a slightly deflected position of the bar. Consideration

nontrivial solution, we must have

$$P = P_{cr} = k_s L \quad (1.4.11)$$

Energy Approach

The strain energy of the system at its deflected state is equal to the strain energy stored in the structural spring.

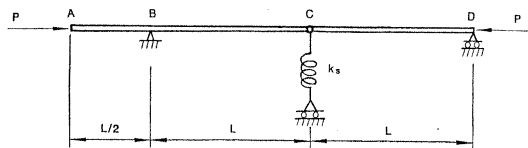
$$U = \frac{1}{2} k_s (L\theta)^2 \quad (1.4.12)$$

The potential energy of the system is equal to the negative of the work done by the applied force.

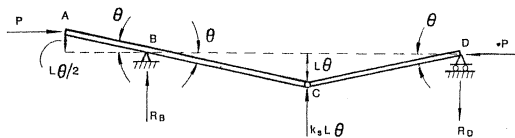
$$V = -PL(1 - \cos \theta) \quad (1.4.13)$$

The total potential energy is

$$\Pi = U + V = \frac{1}{2} k_s (L\theta)^2 - PL(1 - \cos \theta) \quad (1.4.14)$$



(a)



Summing moment about C for the free body in Fig. 1.13c gives

$$R_D L - PL\theta = 0 \quad (1.4.20)$$

Upon substitution of Eq. (1.4.19) into Eq. (1.4.20), we obtain

$$\frac{1}{2}k_s L\theta - \frac{2}{3}P\theta = 0 \quad (1.4.21)$$

The nontrivial solution for the equilibrium equation (1.4.21) gives the critical load of the system as

$$P = P_{cr} = \frac{3}{2}k_s L \quad (1.4.22)$$

Energy Approach

The strain energy of the system is

$$U = \frac{1}{2}k_s (L\theta)^2 \quad (1.4.23)$$

The potential energy of the system is

1.4.4 Three-Bar System

Figure 1.14a shows a three-bar system supported at the ends A and D by frictionless hinges and connected to one another at B and C by linear rotational springs of spring stiffness k_s . The system is assumed geometrically perfect in that the springs are unstrained when all the bars are in their horizontal orientation. We shall now determine the critical load P_{cr} of the system.

Bifurcation Approach

In using the bifurcation approach, we are investigating the equilibrium conditions of the system in a slightly deflected configuration. A configuration of the system displaced by an arbitrary kinematic admissible displacement is shown in Fig. 1.14b. A *kinematic admissible displacement* is a displacement that does not violate the constraints of the system. Note that the two parameters (θ_1 and θ_2) are necessary to describe fully the displaced configuration of the bars. The equilibrium

1.4 Illustrative Examples—Small Deflection Analysis

A and D are zeros. By using the free body diagrams of the left and right bars, respectively (Fig. 1.14c), we can write the following equilibrium equations:

$$k_s(2\theta_1 - \theta_2) - P(L\theta_1) = 0, \quad (1.4.30a)$$

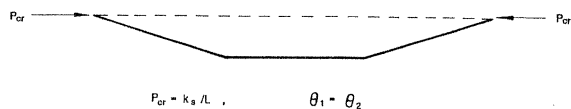
$$k_s(2\theta_2 - \theta_1) - P(L\theta_2) = 0. \quad (1.4.30b)$$

In matrix form, we have

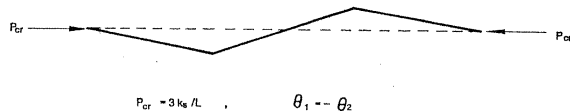
$$\begin{bmatrix} 2k_s - PL & -k_s \\ -k_s & 2k_s - PL \end{bmatrix} \begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \quad (1.4.31)$$

For a nontrivial solution, we must have the determinant of the coefficient matrix equal to zero

$$\det \begin{vmatrix} 2k_s - PL & -k_s \\ -k_s & 2k_s - PL \end{vmatrix} = (2k_s - PL)^2 - k_s^2 = 0 \quad (1.4.32)$$



(a) Symmetric Buckling Mode



$$\begin{aligned} \frac{\partial \Pi}{\partial \theta_2} &= -k_s(2\theta_1 - \theta_2) + 2k_s(2\theta_2 - \theta_1) \\ &\quad - PL[\sin \theta_2 - \sin(\theta_1 - \theta_2)] = 0 \end{aligned} \quad (1.4.39b)$$

Upon simplification and using small angle approximation, we can write Eqs. (1.4.39a and b) in matrix form as

$$\begin{bmatrix} 5k_s - 2PL & -4k_s + PL \\ -4k_s + PL & 5k_s - 2PL \end{bmatrix} \begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (1.4.40)$$

For nontrivial solution, we must have

$$\det \begin{vmatrix} 5k_s - 2PL & -4k_s + PL \\ -4k_s + PL & 5k_s - 2PL \end{vmatrix} = 0 \quad (1.4.41)$$

from which the two eigenvalues are

The equilibrium is stable if all of the following conditions are satisfied:

$$\frac{\partial^2 \Pi}{\partial \theta_1^2} > 0 \quad (1.4.45a)$$

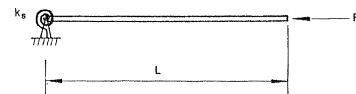
$$\frac{\partial^2 \Pi}{\partial \theta_2^2} > 0 \quad (1.4.45b)$$

$$\left(\frac{\partial^2 \Pi}{\partial \theta_1^2} \right) \left(\frac{\partial^2 \Pi}{\partial \theta_2^2} \right) > \left(\frac{\partial^2 \Pi}{\partial \theta_1 \partial \theta_2} \right)^2 \quad (1.4.45c)$$

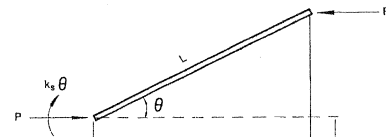
In view of Eqs. (1.4.44a–c), Equations (1.4.45a–c) become

$$P < \frac{5}{2} \left(\frac{k_s}{L} \right) \quad (1.4.46a)$$

$$P < \frac{5}{2} \left(\frac{k_s}{L} \right) \quad (1.4.46b)$$



(a)



The strain energy of the system is equal to the strain energy of the spring

$$U = \frac{1}{2}k_s\theta^2 \quad (1.5.1)$$

The potential energy of the system is the potential energy of the external force

$$V = -PL(1 - \cos \theta) \quad (1.5.2)$$

The total potential energy of the system is then

$$\Pi = U + V = \frac{1}{2}k_s\theta^2 - PL(1 - \cos \theta) \quad (1.5.3)$$

For equilibrium, we must have

$$\frac{d\Pi}{d\theta} = k_s\theta - PL \sin \theta = 0 \quad (1.5.4)$$

This equilibrium equation is satisfied for all values of P if $\theta = 0$. This

postbuckling path is given by

$$P = \frac{k_s\theta}{L \sin \theta} \quad (1.5.5)$$

This path is also plotted in Fig. 1.17 for the range $-\pi/2 \leq \theta \leq \pi/2$. It intersects the fundamental path at $P_{cr} = k_s/L$.

To study the stability of the equilibrium paths, we need to examine the higher order derivatives of the total potential energy function. By taking the second derivative of Π , we have

$$\frac{d^2\Pi}{d\theta^2} = k_s - PL \cos \theta \quad (1.5.6)$$

For the fundamental path, $\theta = 0$, therefore Eq. (1.5.6) becomes

$$\frac{d^2\Pi}{d\theta^2} = k_s - PL \quad (1.5.7)$$

thus the first nonzero term of the series is the fifth term

$$\left. \frac{1}{24} \frac{d^4 \Pi}{d\theta^4} \right|_{\theta=0} \theta^4 = \frac{1}{24} PL\theta^4 \quad (1.5.10)$$

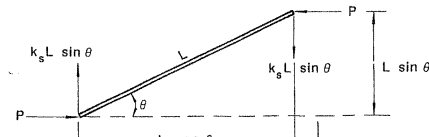
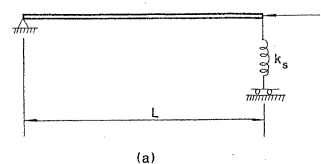
At $P = P_{cr} = k_s/L$, we have

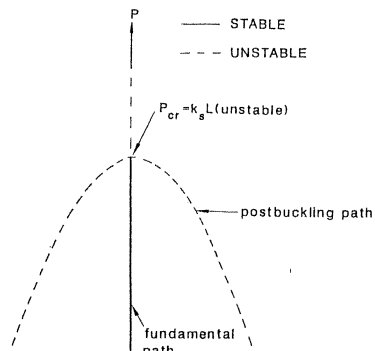
$$\frac{1}{24} PL\theta^4 = \frac{1}{24} k_s \theta^4 \quad (1.5.11)$$

which is positive, indicating that Π is a local minimum and so the equilibrium state at the critical point is stable.

From the above discussion it can be seen that as P increases gradually to P_{cr} , the stable fundamental equilibrium path bifurcates into an unstable equilibrium path corresponding to the original horizontal position of the bar and a stable postbuckling equilibrium path corresponding to the deflected configuration of the bar. The equilibrium state at the critical point is stable. The stable postbuckling equilibrium

1.5 Illustrative Examples—Large Deflection Analysis





1.6 Illustrative Examples—Imperfect Systems

At $P = P_{cr} = k_s L$, we have

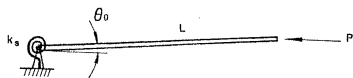
$$\frac{1}{24}(-4k_s L^2 + PL)\theta^4 = -\frac{1}{8}k_s L^2 \theta^4 \quad (1.5.21)$$

which is negative, indicating that Π is a local maximum and so the equilibrium state at the critical point is unstable.

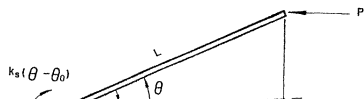
From the above discussion, it can be seen that as P increases gradually to P_{cr} , the stable fundamental equilibrium path bifurcates into an unstable equilibrium path corresponding to the original horizontal position of the bar and an unstable equilibrium path corresponding to the deflected configuration of the bar. The equilibrium state at the critical point is also unstable.

1.6 ILLUSTRATIVE EXAMPLES—IMPERFECT SYSTEMS

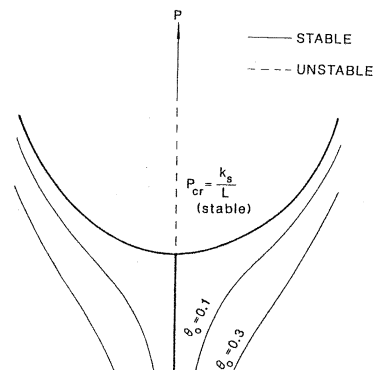
General Principles



(a)



1.6 Illustrative Examples—Imperfect Systems



and they are unstable if

$$P > \frac{k_s}{L \cos \theta} \quad (1.6.8)$$

It is quite obvious that Eq. (1.6.7) always controls for $-\pi/2 \leq \theta \leq \pi/2$. Therefore, the equilibrium paths described by Eq. (1.6.5) are always stable, indicating that as P increases the deflections increase, as shown by the curves with the initial angle $\theta_0 = 0, 0.1$ and 0.3 , and no instability will occur. The maximum load that the system can carry is greater than the critical load, P_{cr} .

1.6.2 Rigid Bar Supported by a Translation Spring

Consider now the imperfect system shown in Fig. 1.22. The system is imperfect, being tilted by an angle θ_0 when the spring is unstretched. We shall now use the energy approach to study the occurrence of this

Energy Approach

The strain energy of the system is equal to the strain energy stored in the spring

$$U = \frac{1}{2} k_s L^2 (\sin \theta - \sin \theta_0)^2 \quad (1.6.9)$$

The potential energy of the system is equal to the potential energy of the external force

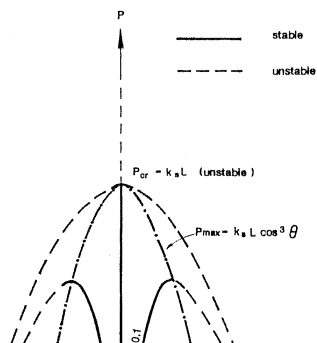
$$V = -PL(\cos \theta_0 - \cos \theta) \quad (1.6.10)$$

The total potential energy of the system is

$$\begin{aligned} \Pi = U + V &= \frac{1}{2} k_s L^2 (\sin \theta - \sin \theta_0)^2 \\ &\quad - PL(\cos \theta_0 - \cos \theta) \end{aligned} \quad (1.6.11)$$

For equilibrium, we must have

$$\frac{d\Pi}{d\theta} = 0$$



and they will be unstable if

$$\sin \theta_0 < \sin^3 \theta \quad (1.6.20)$$

So, it can be seen that the rising equilibrium paths are stable and the falling equilibrium paths are unstable. As the load P is increased from zero, the load-deflection behavior of the imperfect system will follow the stable rising equilibrium paths as shown in Fig. 1.23, until $P_{\max}$ is reached, after which the system will become unstable.

From the foregoing examples of imperfect systems, we can conclude that for a system with stable postbuckling equilibrium paths, a small imperfection will not significantly affect the system's behavior. The maximum load the imperfect system can carry is larger than the critical load of the perfect system. On the other hand, for a system with unstable postbuckling equilibrium paths, a small imperfection may have a notice-

where

R_n = nominal resistance of the structural member expressed in unit of stress

Q_n = nominal working, or service stresses computed under working load conditions

$F.S.$ = factor of safety

i = type of load (i.e., dead load, live load, wind load, etc.)

m = number of load type

The left hand side of Eq. (1.7.1) represents the allowable stress of the structural member or component under various loading conditions (for example, tension, compression, bending, shear, etc.). The right hand side of the equation represents the combined stress produced by various load combinations (for example, dead load, live load, wind load, etc.). Formulas for the allowable stresses for various types of structural

Note that in plastic design, safety is incorporated in the load term and is evaluated at the ultimate (plastic strength) limit state.

1.7.3 Load and Resistance Factor Design

The purpose of load and resistance factor design (LRFD) is to ensure that the nominal resistance of the structural member or component exceeds that of the load effects. Two safety factors are used, one applied to the loads, and the other to the resistance of the materials. Thus, the load and resistance factor design has the format

$$\phi R_n \geq \sum_{i=1}^m \gamma_i Q_{ni} \quad (1.7.3)$$

where

R_n = nominal resistance of the structural member

Q_n = load effect (e.g., axial force, shear force, bending moment,

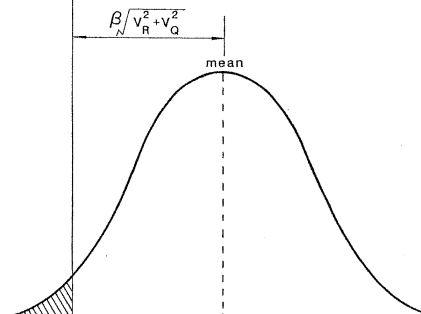
Table 1.1 Load Factors and Load Combinations^a

1.4 D
1.2 $D + 1.6 L + 0.5 (L_r \text{ or } S \text{ or } R)$
1.2 $D + 1.6 (L_r \text{ or } S \text{ or } R) + (0.5 L \text{ or } 0.8 W)$
1.2 $D + 1.3 W + 0.5 L + 0.5 (L_r \text{ or } S \text{ or } R)$
1.2 $D + 1.5 E + (0.5 L \text{ or } 0.2 S)$
0.9 $D - 1.3 W \text{ or } 1.5 E$

where

D = dead load
 L = live load
 L_r = roof live load
 W = wind load
 S = snow load
 E = earthquake load
 R = nominal load due to initial rainwater or ice ex-

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to stiffness of the structure. One important criterion related to stiffness is that the structure or structural component must not deflect excessively under service load conditions. Thus, regardless of the design method, one should always investigate such *serviceability requirements* as deflection and vibration at service load conditions.

1.8 SCOPE

In this book, the discussion of *stability theory* will be limited to conservative systems under static or quasistatic loads. A *conservative system* is a system that is subjected only to conservative forces, that is, to forces whose potential energy is dependent only on the final values of deflection. In addition, most of the material presented in this book's later chapters will be based on what is called *small displacement theory*. Hence, the critical load but not the postbuckling behavior of the member or structure will be studied. Only the stability behavior of structural

References

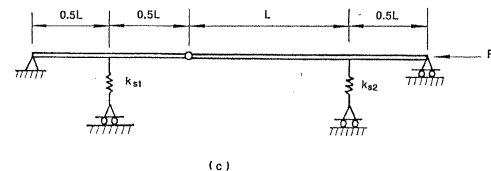


FIGURE P1.1

1.2 Repeat Prob. 1.1, using the energy approach.

1.3 Investigate the stability behavior of the asymmetric spring-bar model shown in Fig. P1.3.

